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SYSTEMATIC REVIEW

# Does Maximal Strength or Speed-Strength Have Stronger Associations with Performance Indicators in the Rugby Football Codes? A Systematic Review and Meta-analysis

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## Abstract

**Background** Both maximal strength and speed-strength are considered key aspects of rugby league and rugby union match-play yet represent distinct physical qualities. Establishing whether maximal strength or speed-strength has a greater association with game performance in these sports can help direct resistance training interventions towards the physical quality most likely to transfer to the outcome of interest. It is therefore important to develop a clearer understanding of whether it is maximal strength or speed-strength that has stronger links to key aspects of rugby union and rugby league competition.

**Objectives** To systematically review and meta-analyse the literature to compare maximal strength versus speed-strength measures in their associations with within-game key performance indicators (KPIs) in rugby union and rugby league. Further, sub-group analysis was undertaken to determine whether the type of KPI (tackles, rucks, line breaks, carries/hit-ups, errors, and miscellaneous) moderated the extent to which the observed effect favoured maximal strength or speed-strength for a given KPI.

**Methods** A systematic search, conducted in accordance with PRISMA guidelines, of Web of Science, PubMed, and SportDiscus was conducted up to December 2024. Studies meeting the inclusion criteria provided correlation coefficients that were converted into Fisher's z-transformed effect sizes to standardise the measures and allow for consistent comparison across studies. These effect sizes were analysed using a multivariate meta-analysis framework.

**Results** Six studies containing 41 maximal strength versus speed-strength comparisons and 134 players were included in the final analysis. Of the 786 studies initially identified, 28 underwent full-text screening, with 22 ultimately excluded. The overall pooled effect was 0.24 [0.11: 0.37] in favour of strength; however, the prediction interval ranged from $g = -0.58$ to $g = 1.06$. The moderator analysis revealed a statistically significant effect of type ($p = 0.038$), indicating stronger associations for maximal strength compared to speed-strength. Residual heterogeneity ($p = 0.005$) suggested substantial variability in effect sizes across studies that was not fully explained by the model.

**Conclusions** Within-game rugby union and rugby league KPIs typically have stronger links to maximal strength than to speed-strength. However, the wide prediction interval underscores that the observed maximal strength advantage may not generalise across all types of KPIs or predictably translate to all future settings.

## 1 Introduction

Both maximal strength [1] and speed-strength [2] are considered key aspects of sports performance yet represent distinct physical qualities [3, 4]. Maximal strength refers to both the greatest load lifted for one or few repetitions (e.g., a 1, 3 or 5 repetition maximum (RM) back squat) and the greatest amount of force produced during a maximal isometric task (e.g., peak force derived from an isometric mid-thigh pull (IMTP) test) [3]. Conversely, speed-strength describes unloaded or light loaded (up to approximately + 60% body mass) actions performed with the goal of maximising velocity at take-off or release (e.g., a countermovement jump (CMJ), commonly referred to as fast strength or 'power'), and the ability to produce force in a fast (< 0.250 s) stretch shortening cycle (SSC) action (e.g., a drop jump (DJ), commonly referred to as reactive strength). Maximal strength and speed-strength measures typically demonstrate 10–40% shared variance and therefore require specific training and testing methods [3], particularly as an individual becomes more trained [5, 6]. Decisions regarding when to focus on maximal strength versus speed-strength training are a key

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**Key Points**

Maximal strength is a primary correlate of key in-game actions in rugby league and rugby union, particularly in actions like tackling and error minimisation. Speed-strength plays a complementary role, contributing to KPIs such as carries and line breaks, though its influence is less pronounced than that of maximal strength.

The relationship between physical qualities and KPIs is context-dependent, influenced by factors such as game models, competition levels, and player profiles.

These findings can be considered when designing resistance training interventions targeted towards specific aspects of rugby league and rugby union match play.

aspect of an athlete's physical preparation, particularly in sports where expressions of both qualities occur during competition. A notable example of this is in rugby union and rugby league, which are team-based, field-invasion sports, played internationally at the amateur and professional level. Rugby union features 15 players per side, contested scrums and lineouts, and longer phases of play, while rugby league is played with 13 players, has simpler restarts, and places greater emphasis on repeated high-intensity efforts and structured set-play cycles. From a physical perspective these sports are characterised by high-speed collisions, tackling (wrestling and grappling), repeated sprinting, and jumping [7, 8]. Success in these sports is a complex interaction of players within and between teams, positional demands, game style tactics, and physical qualities [9, 10]. Some aspects of performance in rugby union and rugby league may reflect maximal strength, whist others are underpinned by speed-strength. Thus, understanding game actions and their impact on competition outcome, in addition to the physical qualities relevant to those actions, is necessary to inform training and coaching practices.

In a sporting context, key performance indicators (KPIs) are used to describe important aspects of match play [11]. These indicators are typically an action variable associated with successful performance or competition outcome and provide a mechanism by which the complexity of competition can be quantified and therefore considered in the training process [11, 12]. For example, in the rugby codes, KPIs may capture metrics related to tackles, carries, scrum, ruck, and errors [7, 8]. Technical, skill-based training can then be targeted towards the features of gameplay represented by the KPI of interest. However, examining KPIs in isolation may provide limited insight into how these actions can be improved [13]. For example, it is evident that reduced errors and increased dominant tackles are desirable, but without knowledge of the modifiable factors associated with these metrics, improving these KPIs through training is difficult.

One approach to operationalising KPI information within the training process is via exploring their links to physical qualities, including maximal strength and speed-strength. Such an approach considers that a player's technical performance is influenced to some extent by their physical development. Correlations between KPIs and maximal strength and speed-strength measures have been successfully revealed across rugby codes, ranging from trivial to large, depending on the test and KPI of interest [14–16]. However, statistical comparison between coefficients in this context is uncommon, which means it not possible to know if a significant difference is present between two correlations (e.g., a maximal strength vs. speed-strength coefficient with respect to a given KPI). Furthermore, individual studies in these sports are limited by sample size given the number of players on a team is inherently capped, which results in unstable correlation estimates and reduced precision [17]. Another limitation of examining such studies in isolation is that the results may not be generalisable to other contexts [18]. This is particularly the case in team sports like rugby union and rugby league where game models and playing styles vary across teams within a competition. For example, some teams prioritise structured set-piece dominance or high-contact phases, while others favour unstructured play, rapid ball movement, or width-based attack. These tactical differences influence not only the frequency and type of KPIs observed, but also the physical qualities required to execute them effectively.

Establishing whether maximal strength or speed-strength has a greater association with within-game performance in rugby union and rugby league can help inform the relative emphasis placed on each within a resistance training plan. For example, speed-strength may erroneously be targeted, resulting in a potential decrease in strength and negative transfer to on field performance. This knowledge enables practitioners to make more informed decisions about programming and testing foci, reducing the risk of misallocated training time and improving the likelihood of meaningful transfer to match-play [19]. It is therefore important to develop a clearer understanding of whether it is maximal strength or speed-strength that has stronger links to key aspects of rugby union and rugby league performance.

Consequently, the purpose of this investigation was to systematically review and meta-analyse the literature to determine whether maximal strength versus speed-strength measures had stronger associations with within-game KPIs in rugby union and rugby league. Further, sub-group analysis was undertaken to determine whether the type of KPI (e.g., tackles, rucks, line breaks, carries/hit-ups, errors) had an impact on the effect.

## 2 Methods

### 2.1 Search Strategy

Following the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework, a systematic search was conducted in December 2024 across the databases Web of Science, SPORTDiscus, and PubMed. The search targeted peer-reviewed original research articles published in English. The full search string was: ("performance indicators" OR "KPI" OR "match-play" OR "game-play" OR "outcomes" OR "match") AND ("strength") AND ("Rugby"). Additionally, the reference lists of the included studies were manually reviewed to identify further relevant articles. Titles, abstracts and full texts were independently screened by two reviewers (LJ and VK). The study protocol was registered with INPLASY (2024120031).

### 2.2 Selection Criteria

Search results were imported into EndNote (X9, Thomson Reuters) to facilitate the removal of duplicates and further organisation. Titles and abstracts were screened for relevance followed by a full-text review of the remaining articles. Studies were included if they met the following criteria: (1) they reported correlation coefficients between within-game KPIs and measures of maximal strength and speed-strength in rugby league or rugby union; (2) a correlation coefficient between both a maximal strength and a speed-strength measure in relation to the same KPI was reported; (3) cohorts were of adult-level competition (i.e., not age restricted); (4) they were published in a peer-reviewed journal before November 2024; and (5) the full text was available in English. Studies that did not meet these criteria were excluded. Data were extracted on authorship, KPI type, group characteristics, maximal strength and speed-strength measures, correlation coefficients and associated p-values. To allow sub-group analysis, KPIs were distributed into the following categories: tackles, errors, carries/hit-ups, line breaks, rucks and miscellaneous. KPI categories were derived based on the variables most consistently reported across studies and grouped according to their similarity in match-play demands.

### 2.3 Selection of Outcome Variables

Based on Hughes and Bartlett's [11] definition, KPIs were considered action variables closely tied to successful performance or competition outcome. With regard to maximal strength measures, these included 1–5RM scores for primary lower-body lifts (e.g., back squat, deadlift, power clean) and peak force derived from lower-body isometric tests such as the isometric squat or IMTP. Speed-strength outcomes included measures obtained from the CMJ, SJ, broad jumps, drop jumps (fast SSC), depth jumps (slow SSC), and repeated hop tests. In cases where multiple tests of maximal strength (e.g., 1RM back squat and peak isometric mid-thigh pull force) or speed-strength (e.g., CMJ and SJ) were reported, the test with the strongest association was used. Where multiple measures of speed-strength were reported (e.g., jump height, peak power, peak force), the biomechanically resultant (i.e., output) measure was prioritised. Therefore, speed-strength metric selection priority was (1) jump height, (2) take-off velocity, (3) peak velocity, (4) peak power. Missing or unclear data were assumed unreported or irrelevant; authors were contacted for clarification where possible.

### 2.4 Assessment of Reporting Quality

To assess the methodological quality of the included studies, a modified version of the Downs and Black Quality Index tool [20] was applied, consistent with previous research practices [21–23]. For the purposes of this review, ten items from the checklist were deemed relevant (see Electronic Supplementary Material 1), with questions pertaining to patient treatment, training interventions, and group randomization excluded due to their lack of relevance to the research focus [23]. Each item was scored as 1 (yes) or 0 (no /unable to determine), yielding a maximum possible score of 10. The reporting quality assessment was conducted by the first author of the present investigation (LJ).

### 2.5 Statistical Analysis

To examine the relationship between study effect sizes and their moderators, we conducted a multivariate meta-analysis using the *metafor* package in R [24]. Effect sizes for each study were calculated as Fisher's z-transformed correlation coefficients (ZCOR) using the *escalc* function. Fisher's z-transformation was chosen because it standardises the correlation coefficients, allowing for the combination of both Pearson and Spearman correlations, which were reported in different studies. For each effect size, we used the reported correlation coefficients (ri) and sample sizes (ni) from the studies. To account for the dependency among multiple effect sizes reported within the same study, we estimated a block-diagonal variance–covariance matrix using the *clubSandwich* package in R [25]. A conservative average correlation of 0.5 among effect sizes within each study was assumed.

A random-effects multivariate meta-analysis model was then fitted, which included a random intercept for each study

and nested random effects for each effect size within the studies. The model allowed us to estimate the variability both between studies and within studies (i.e., the variability of effect sizes within each study). The moderator variable type (which categories effects into either maximal strength or speed-strength) was included as a fixed effect to examine its influence on the study effect sizes. The overall model results, including the estimated variance components and moderator effects, were interpreted to understand the factors contributing to the variation in effect sizes across studies.

To provide a more comprehensive understanding of the uncertainty in effect size estimates, we also calculated prediction intervals, which indicate the expected range of true effect sizes in a new study. Unlike confidence intervals, which reflect the precision of the estimated mean effect, prediction intervals incorporate both within-study and between-study variance, offering insight into the expected dispersion of future observations.

## 3 Results

### 3.1 Search Results

Following the removal of duplicates, 786 studies were retained. Of these, 28 articles progressed to full-text screening after the initial title and abstract review. Upon completing the full-text assessment, 11 articles met the inclusion criteria and no disagreements occurred between the two authors screening the articles. However, five articles were excluded as relevant correlation coefficients were not reported, leaving six studies included in the meta-analysis and systematic review [15, 16, 26–29]. The PRISMA framework used in this process is illustrated in Fig. 1. The eligible studies generated 41 comparisons. A total of 134 participants (rugby union = 77; rugby league = 57) were included, representing cohorts of semi-professional ($n = 43$), sub-elite ($n = 63$), professional ($n = 14$), and international-level ($n = 14$) players.

### 3.2 Outcome of Assessment of Reporting Quality

Study quality assessment scores (Table 1) ranged from 8 to 9 on the 10-point scale, with a mean of 8.67 (± 0.51), indicating a high methodological quality among the included studies based on this metric. Question seven was not met in any of the studies, meaning external validity cannot be confirmed based on this assessment.

### 3.3 Study Characteristics

Tackles were analysed in 15 comparisons across three studies, while errors were examined in nine comparisons from three studies. Carries or hit-ups were featured in five comparisons across three studies, and line breaks were assessed in five comparisons across two studies. Rucks were analysed in two comparisons from two studies, and miscellaneous measures were included in five comparisons across three studies. Miscellaneous KPIs included tries scored, play-the-ball speed, 'marker' (player from marker position makes first tackle attempt (%)), and off-load.

The 1RM back squat test served as a strength measure in 40 comparisons, with peak force from the isometric mid-thigh pull (IMTP) used in one comparison. Speed-strength tests included the countermovement jump (CMJ), which was represented in 37 comparisons, the broad jump, featured in three comparisons, and the drop jump, which was used in one comparison. Table 2 provides an overview of the studies included in the analysis.

### 3.4 Meta-Analysis

A multivariate meta-analysis was conducted using a restricted maximum likelihood (REML) estimation method to evaluate the effect sizes across studies and the potential moderating effect of type on the effect sizes. The model included 41 effect sizes across the studies. Forest plots of these Fisher's z-transformed effects, clustered by the study KPIs, are shown in Fig. 2.

The log-likelihood of the model was -18.97 with an Akaike Information Criterion (AIC) of 53.94 and a Bayesian Information Criterion (BIC) of 66.38. The residual heterogeneity was statistically significant, $QE\ (df = 35) = 60.25$, $p = 0.005$, indicating that there is substantial variability in effect sizes that is not accounted for by the model. The estimated variance components suggested that there was small variance between studies ($\sigma_1^2 = 0.029$), while the variance within studies (i.e., among effect sizes within the same study) was $\sigma_2^2 = 0.059$.

A second model was fitted to test for the effect of type as a moderator. This model found the effect of type to be statistically significant ($Q_M(1) = 4.32$, $p = 0.038$), suggesting that type does moderate the effect sizes in this study (see Fig. 3), with the effect size being larger, on average, for maximal strength compared to speed-strength.

## 4 Discussion

The objective of this review and meta-analysis was to determine whether maximal strength or speed-strength measures had stronger associations with within-game KPIs in rugby union and rugby league. This study also conducted sub-group analyses to determine whether the type of KPI (e.g., tackles, rucks, line breaks, carries/hit-ups, errors) influenced the effect. Six studies were included in the final review, and analysis consisted of 41 maximal

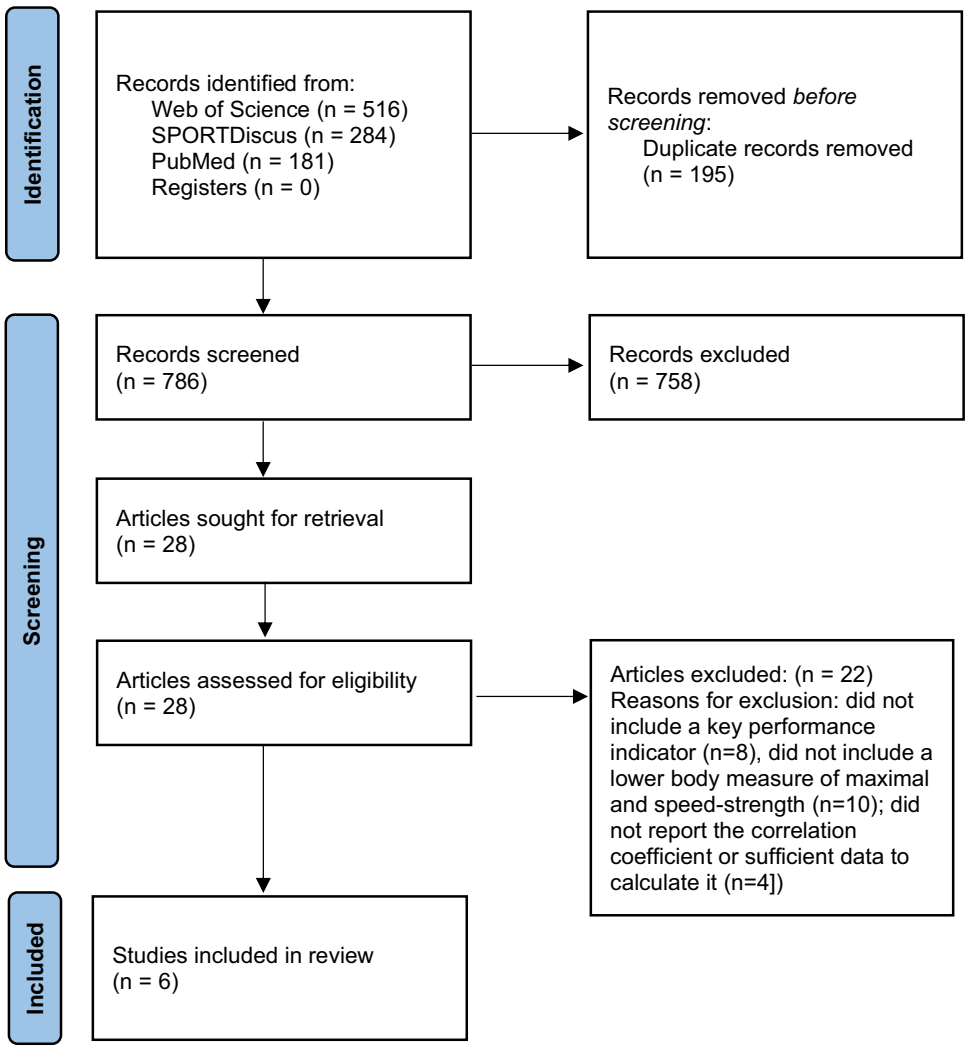


**Fig. 1** PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) flow diagram describing the selection processes for eligible articles

**Table 1** Results of study quality assessment

| Study | Downs and Black checklist item number | | | | | | | | | | Total score/10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Reporting | | | | | | External validity | Internal validity | | | |
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | |
| Coghetto et al., 2023 [26] | + | + | + | + | + | + | ? | + | + | + | 9 |
| Cunningham et al., 2018 [15] | + | + | + | + | + | − | ? | + | + | + | 8 |
| Hart et al., 2023 [27] | + | + | + | + | + | + | ? | + | + | + | 9 |
| Redman et al., 2022 [28] | + | + | + | + | + | + | ? | + | + | + | 9 |
| Speranza et al., 2015 [16] | + | + | + | + | + | + | ? | + | + | + | 9 |
| Johnston, 2019 [29] | + | + | + | + | − | + | ? | + | + | + | 8 |

strength versus speed-strength comparisons. The overall finding of this systematic review and meta-analysis demonstrated that maximal strength had significantly stronger associations with competition KPIs ($g = 0.25$, 95% CI [0.02, 0.47]). However, wide prediction intervals were present, indicating that the results may not generalise across all types of KPIs or predictably translate across all future settings. Improvements in KPIs related to tackles and errors were clearly associated with greater maximal strength versus speed-strength levels, while line breaks, carries and rucks were equally influenced by maximal strength and speed-strength. These findings can help the

**Table 2** Study characteristics

| Study | *n* | Matches | Population | KPI label | KPI type | Maximal strength metric | Maximal strength correlation coefficient | Speed-strength Metric | Speed-strength Correlation Coefficient |
|---|---|---|---|---|---|---|---|---|---|
| Coghetto et al., 2023 [*ρ*] [26] | 25 | 12 | Sub-elite rugby union players | Successful Carries | Carries/Hit Ups | 1RM Squat | −0.25 | Broad Jump Distance | 0.49 |
| | | | | Successful Attacking Rucks | Rucks | 1RM Squat | −0.17 | Broad Jump Distance | 0.05 |
| | | | | Successful Tackles | Tackles | 1RM Squat | 0.11 | Broad Jump Distance | 0.13 |
| Cunningham et al., 2018 (Backs) [*r*] [15] | 14 | 92 | International level rugby union players | Effective Attacking Ruck (%) | Rucks | IMTP PF | 0.63 | DJ RSI | 0.58 |
| Hart et al., 2023 (Backs) [*r*] [27] | 16 | 22 | Sub-elite rugby union players | Dominant Collisions | Carries/Hit Ups | 1RM Squat/BM | 0.73 | CMJ Height | −0.23 |
| | | | | Handling Turnovers | Error | 1RM Squat/BM | 0.48 | CMJ Height | 0.10 |
| | | | | Total Turnovers | Error | 1RM Squat/BM | 0.54 | CMJ Height | 0.18 |
| | | | | Line Breaks | Line Breaks | 1RM Squat/BM | 0.43 | CMJ Height | 0.01 |
| | | | | Defenders Beaten | Line Breaks | 1RM Squat/BM | 0.12 | CMJ Height | −0.04 |
| | | | | Tries Scored | Miscellaneous | 1RM Squat/BM | 0.53 | CMJ Height | 0.08 |
| | | | | Tackle Breaks | Tackles | 1RM Squat/BM | 0.73 | CMJ Height | −0.22 |
| Hart et al., 2023 (Forwards) [*r*] [27] | 22 | 22 | | Dominant Collisions | Carries/Hit Ups | 1RM Squat/BM | 0.43 | CMJ Height | 0.24 |
| | | | | Handling Turnovers | Error | 1RM Squat/BM | 0.42 | CMJ Height | 0.21 |
| | | | | Total Turnovers | Error | 1RM Squat/BM | 0.44 | CMJ Height | 0.15 |
| | | | | Line Breaks | Line Breaks | 1RM Squat/BM | 0.33 | CMJ Height | 0.28 |
| | | | | Defenders Beaten | Line Breaks | 1RM Squat/BM | 0.32 | CMJ Height | 0.35 |
| | | | | Tries Scored | Miscellaneous | 1RM Squat/BM | 0.45 | CMJ Height | 0.42 |
| | | | | Tackle Breaks | Tackles | 1RM Squat/BM | 0.42 | CMJ Height | 0.44 |
| Redman et al., 2022 [*ρ*][28] | 14 | 5 | Professional rugby league players | Post-contact meters, m | Carries/Hit Ups | 1RM Back Squat | 0.42 | CMJ Height | −0.37 |
| | | | | Meters conceded post-contact, m | Carries/Hit Ups | 1RM Back Squat | −0.39 | CMJ Height | −0.31 |
| | | | | Line breaks, % | Line Breaks | 1RM Back Squat | −0.35 | CMJ Height | 0.29 |

**Table 2** (continued)

| Study | *n* | Matches | Population | KPI label | KPI type | Maximal strength metric | Maximal strength correlation coefficient | Speed-strength Metric | Speed-strength Correlation Coefficient |
|---|---|---|---|---|---|---|---|---|---|
| | | | | PTB speed, s | Miscellaneous | 1RM Back Squat | −0.54 | CMJ Height | −0.12 |
| | | | | Marker, % | Miscellaneous | 1RM Back Squat | 0.13 | CMJ Height | 0.23 |
| | | | | Broken tackles, % | Tackles | 1RM Back Squat | −0.16 | CMJ Height | 0.37 |
| | | | | Attack tackle to PTB contest won, % | Tackles | 1RM Back Squat | −0.21 | CMJ Height | −0.14 |
| | | | | Attack tackle to PTB contest lost, % | Tackles | 1RM Back Squat | 0.21 | CMJ Height | −0.14 |
| | | | | Effective tackle % | Tackles | 1RM Back Squat | −0.06 | CMJ Height | −0.50 |
| | | | | Ineffective tackle, % | Tackles | 1RM Back Squat | −0.29 | CMJ Height | 0.35 |
| | | | | Dominant tackle, % | Tackles | 1RM Back Squat | 0.54 | CMJ Height | −0.13 |
| | | | | Defense tackle to PTB contest won, % | Tackles | 1RM Back Squat | −0.06 | CMJ Height | −0.23 |
| | | | | Defense tackle to PTB contest lost, % | Tackles | 1RM Back Squat | 0.11 | CMJ Height | 0.43 |
| Speranza et al., 2015 [$\rho$] [16] | 16 | 23 | Semi-professional rugby league players | Error | Error | 1RM Back Squat | −0.39 | CMJ Peak Power | 0.08 |
| | | | | Off-load | Miscellaneous | 1RM Back Squat | −0.32 | CMJ Peak Power | 0.00 |
| | | | | Tackles | Tackles | 1RM Back Squat | 0.27 | CMJ Peak Power | −0.04 |
| | | | | Dominant Tackle | Tackles | 1RM Back Squat | 0.65 | CMJ Peak Power | 0.22 |
| | | | | Positive Tackle | Tackles | 1RM Back Squat | 0.30 | CMJ Peak Power | 0.34 |
| | | | | Missed tackles | Tackles | 1RM Back Squat | −0.23 | CMJ Peak Power | 0.11 |
| Johnston, 2019 (Forwards) [$\rho$] [29] | 15 | 22 | Semi-professional rugby league players | Defensive error rate | Error | 1RM Back Squat/BM | −0.74 | CMJ Peak Power/BM | −0.36 |
| | | | | Attacking error rate | Error | 1RM Back Squat/BM | −0.74 | CMJ Peak Power/BM | −0.30 |
| Johnston, 2019 (Backs) [$\rho$] [29] | 12 | 22 | | Defensive error rate | Error | 1RM Back Squat/BM | −0.64 | CMJ Peak Power/BM | 0.08 |
| | | | | Attacking error rate | Error | 1RM Back Squat/BM | −0.08 | CMJ Peak Power/BM | −0.182 |

*BM* body mass, *CMJ* countermovement jump, *1RM* one repetition maximum, $\rho$ Spearman rank-order correlation coefficient was used, *r* Pearson's correlation coefficient was used

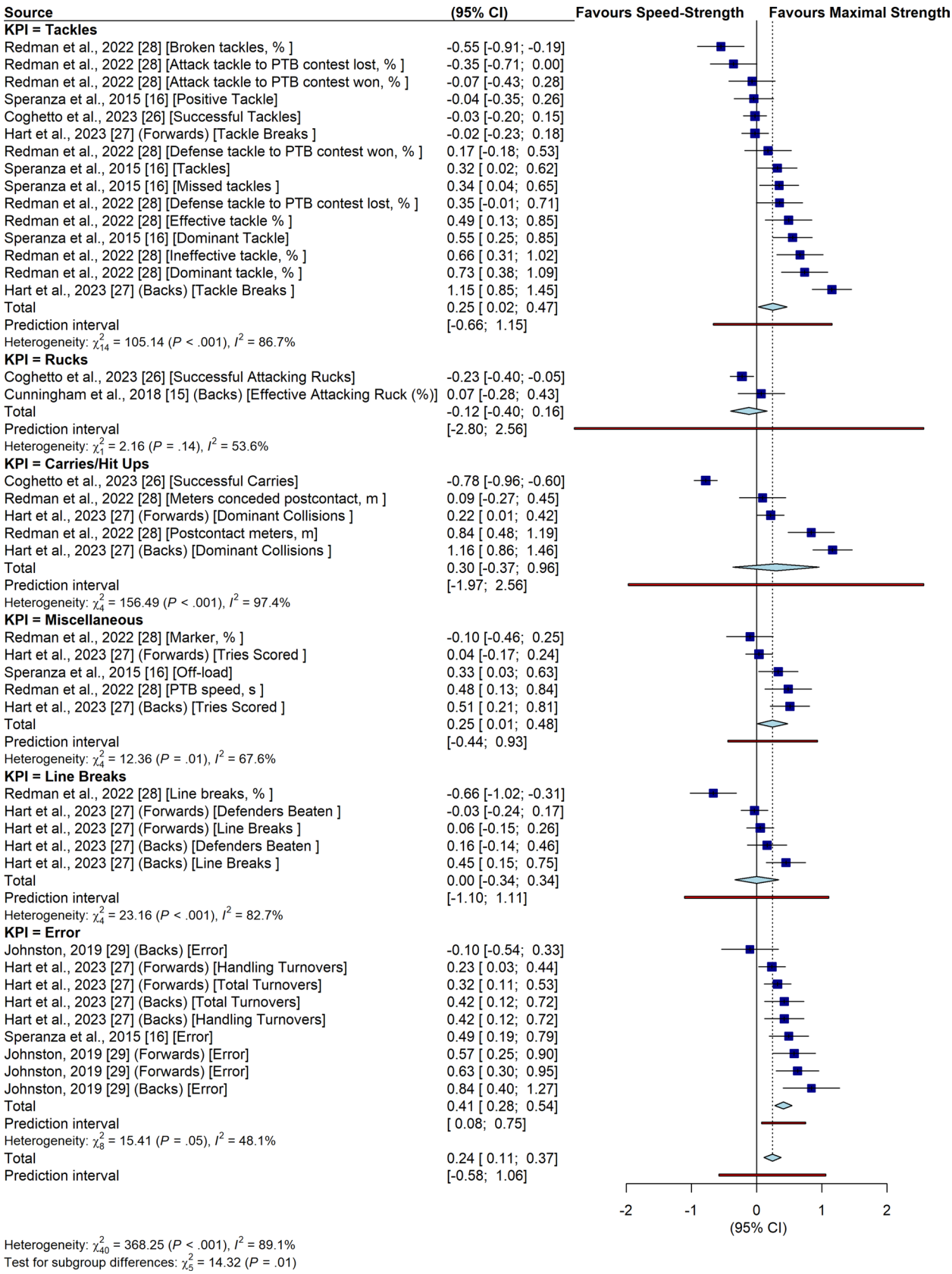


**Fig. 2** Forest plot of the combined studies for the meta-analysis. *CI* confidence interval, *KPI* key performance indicator, *PTB* play the ball

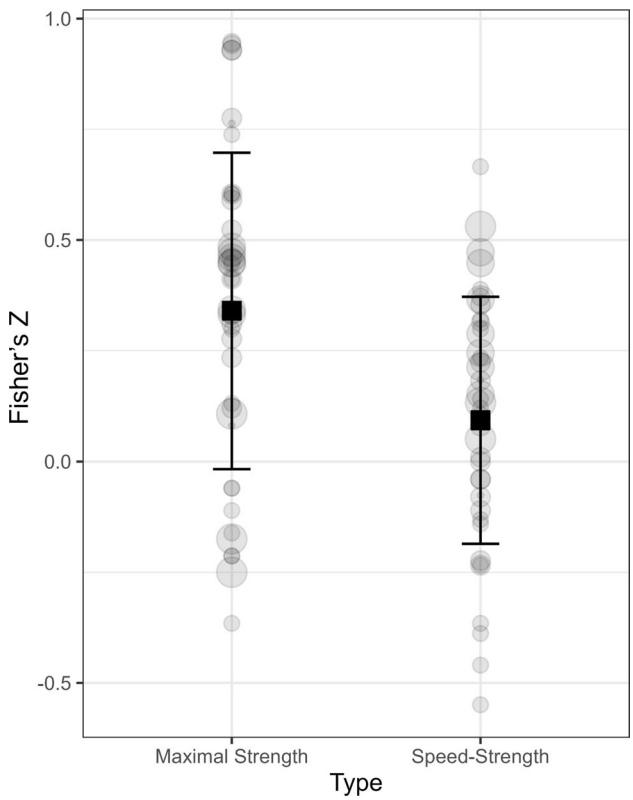


**Fig. 3** A comparison of maximal strength and speed-strength study-level and pooled effects

direction of resistance training interventions towards the aspect of force expression most likely to impact on field performance in the rugby codes.

### 4.1 Overall

Previous reviews have identified both maximal strength and speed-strength as key physical qualities of rugby union [30–32] and rugby league [8, 32] performance; however, it was unclear which of these two qualities are of greater relevance. These findings provide cumulated evidence that maximal strength is a more important physical quality than speed-strength in the context of rugby union and rugby league KPIs. The greater relevance of maximal strength versus speed strength is in alignment with previous investigations that have revealed that back squat strength is a better indicator of playing rank than jump squat barbell peak power [33]. Furthermore, values for maximal strength levels in rugby union and rugby league players are typically among the highest of team sport athletes (1RM back squat ~ 176 kg), yet this is not the case for speed-strength related measures (force platform derived CMJ height = 0.36 m) [34, 35]. While comparisons with lower tier players in the same sport, and to athletes across different sports, provide an indication of the relevance of a physical quality, linking directly to decisive on-field statistics is a more direct link to actual sports performance [19]. From an applied perspective these outcomes suggest that training methods that improve maximal strength are often likely to have a superior transfer to important game actions, particularly in contexts or competition levels where absolute strength capacity is further from a ceiling effect. However, given this analysis measured the *difference* between maximal strength and speed-strength relationships, and strong associations between speed-strength and KPIs in rugby union and rugby league are commonplace [15, 36, 37], the development of speed-strength is still relevant.

Of the 41 comparisons, only four significantly favoured speed-strength (with the upper bound of the confidence interval (CI) not enveloping zero). These four comparisons occurred in two studies [26, 28], yet were distributed across four different KPI types: tackles, rucks, carries/hit-ups and line breaks. The distribution across a range of performance indicators indicates that the relationships may not be reflective of the KPI per se, but rather the team characteristics at the club level. For example, the speed-strength-favouring results in these two studies are likely influenced by contextual factors, including team strategies, player profiles, competition levels, and coaching priorities, rather than inherent differences in the specific KPIs. A team employing an expansive, high-tempo style with frequent offloads and lateral movement may place greater demand on speed-strength qualities, whereas a team built around direct, collision-heavy carries may derive greater benefit from maximal strength. Similarly, rosters composed of lighter, faster backs relative to heavier, stronger forwards will influence the expression and relevance of different strength qualities in KPIs. It is therefore important to consider team-level characteristics when interpreting the relationship between physical qualities and KPIs, as these factors may drive outlier results that do not generalise to other contexts.

### 4.2 Tackles

Tackle performance is an important indicator of success in the rugby codes. For example, tackle skill discriminates across developmental levels [38], while fewer missed tackles are associated with higher ladder position in the National Rugby League (Australia's top professional league) [9]. The meta-analysis results demonstrated a clear and consistent preferential relationship between maximal strength and improved tackle-related KPIs ($g = 0.25$, 95% CI [0.02, 0.47]). This aligns with the biomechanical demands of tackling which requires the ability to produce high forces over relatively long epochs (several seconds) to cause a change in an opponent's momentum [39, 40]. Accordingly, measures of lower body maximal strength have consistently been associated with tackling ability. For example, back squat

strength has demonstrated stronger associations than lower body speed-strength metrics to standardised one-on-one tackling tests in semi-professional rugby league players [41, 42]. Furthermore, improvements in tackle ability have been more strongly related to improvements in strength than to speed-strength [43]. These relationships are likely influenced by baseline strength levels, with larger benefits observed in players further from a potential ceiling effect [5, 6, 44]. At higher levels of maximal strength, additional gains may yield diminishing returns for tackle performance. It is important to note that tackle technique, including body position, timing and contact mechanics, also contributes meaningfully to performance [45], and may differ between rugby league and union due to variations in rules and contest structure. These differences could influence how physical qualities like strength relate to tackle success. Nonetheless, these findings provide strong evidence for the preferential importance of maximal strength versus speed-strength on tackling performance in the rugby codes.

### 4.3 Errors

Of the nine comparisons for the error KPI, eight favoured the strength measure, while one was trivial and non-significant. This demonstrates that an increase in maximal strength has a significantly greater positive impact on errors (i.e., reduction of errors) than speed-strength ($g = 0.41$, 95% CI [0.28, 0.54]). Errors in this context encompassed a range of handling and technical variables, including knock-ons, forward passes, intercepted passes, dropped balls behind, and loss of possession during a tackle. These variables are inherently technical and skill-based, where maximal strength may contribute to maintaining technical proficiency under physical and fatigue-related demands. Furthermore, the magnitude of the effect for the error category was greater than for any other KPI in the analysis. The narrow prediction intervals speak to the consistency of this finding across context including codes (league and union) and positions (forwards and backs). It is possible that skill execution is preserved longer in stronger individuals leading to reduced errors [46]. Further, greater maximal strength improves an athlete's ability to resist external forces and tolerate increases in workloads [47], minimising performance disruption during physical contests [1]. In the present analysis, all nine comparisons for the error KPI were drawn from three studies involving semi-professional or sub-elite teams ($n = 78$ players) across both rugby codes, with data collected in Australia and England. As such, it remains unclear whether the same relationships would be observed at the amateur or the professional level. Nonetheless, the results reveal maximal strength as a more dominant muscle function than maximal speed-strength in reduced errors in rugby league and rugby union.

### 4.4 Carries/Hit-Ups

Somewhat mixed findings were revealed for the carries/hit-ups KPI ($g = 0.30$, 95% CI [–0.37, 0.96]), with three comparisons significantly favouring maximal strength (ranging from a small to a large effect), one significant and large effect towards speed-strength, and the remaining comparison trivial and non-significant. This might be explained by the range of KPI types within this category. For example, the KPIs that were characterized by collisions ('post-contact meters', 'dominant collisions') clearly favoured strength measures. In contrast, the 'successful carries' KPI type is characterised by footwork, posture, passing, and favoured speed-strength to a significant and large extent. These findings are in alignment with previous investigations that have revealed that maximal strength takes primacy over speed-strength in dominating collisions [41–43], whereas this is not necessarily the case for tasks such as change of direction and acceleration [48–50]. It is also likely that positional role contributes to this divergence. Hit-ups are more frequently performed by forwards and are collision-dominant, whereas carries involving evasive movement and ball-handling are more often executed by backs. This distinction may partially explain the mixed findings and suggests that training priorities may differ by position. Although grouped together in this investigation, it may be worthwhile to consider carries and hit-ups as distinct KPIs underpinned by different strength qualities (speed-strength vs. maximal strength).

### 4.5 Line Breaks

The lack of a clear difference between strength and speed-strength associations with line breaks ($g = 0.00$, 95% CI [–0.34, 0.34]) may be explained by the multifaceted, diverse nature of this KPI, which relies on a combination of both maximal strength and speed-strength. Unlike tackle breaks and hit-ups, line breaks are not necessarily characterised by direct physical contact, and primarily occur when a player breaks tackles, which may be linked to maximal strength; as compared with when a player is put through a gap in the defensive line by a pass or is able to evade defenders, which may be more closely linked to qualities such as change-of-direction speed, acceleration, and maximum velocity. As such, measures of speed-strength are likely to have a greater influence on line breaks than other KPIs. This is supported by evidence from rugby union, which indicates that line-break ability

is associated with high-speed expressions [51]. Similar relationships to change-of-direction performance have been reported in strength and speed-strength measures in team sport athletes [48–50] including rugby league [48]. In addition, generally similar improvements in change-of-direction ability occur following maximal strength and speed-strength training [52]. Another key aspect of line breaks is reactive agility whereby a player must change direction or speed in response to an opponent. Relationships between physical qualities such as strength [50, 53] and speed-strength [50, 54–56] to reactive agility performance are typically trivial to small, indicating the technical and sport-specific nature of this aspect of line breaks.

### 4.6 Rucks and Miscellaneous

Only two KPIs across two studies were related to ruck performance, and these demonstrated trivial, non-significant differences between strength versus speed-strength relationships. The relatively small, combined sample ($n = 39$) reduced statistical power and precision in the results, making it challenging to draw meaningful conclusions. The findings from the "miscellaneous" category, encompassing KPIs that did not fit neatly into other defined groups, highlight the nuanced and context-dependent relationship between physical qualities and KPIs in rugby league and rugby union. Overall, the analysis revealed a modest preference for strength ($g = 0.25$, 95% CI [0.01, 0.48]), particularly for the KPIs play-the-ball speed, off-loads, and tries scored by backs. Greater strength levels improve the ability to manage contact [1, 46], which facilitates faster play-the-ball speed following a tackle and maintains control during heavy contact leading to increased passing opportunities (e.g., offloads).

### 4.7 Limitations

There are limitations to the present investigation that should be considered. The meta-analysis design restricted the outcome effect to a scale between only two variables, in this case maximal strength versus speed-strength. However, maximal strength describes two distinct strength qualities – heavy strength and isometric strength, and these were not differentiated in the analysis [3, 57]. This may have masked the specific relevance of heavy dynamic strength, which is more representative of match demands and common training practice [8]. Similarly, speed-strength included both short (< 0.250 s) and longer (0.300 to ~ 1.000 s) SSC actions, despite these representing different types of speed-strength [58]. However, given a short SSC measure (drop jump-derived reactive strength index) featured in only one of the 41 comparisons, any potential impact on the results would be trivial. Furthermore, explosive strength (typically quantified as the rate of force development or instantaneous force during the early part of the isometric force–time curve) was not considered, yet may have relevance to KPIs in rugby union and rugby league [8, 31]. Additionally, the analysis did not distinguish between player positional groups (e.g., forwards vs. backs), competition grade or geographic location. Although this approach allowed for greater statistical power, it may have masked specific associations that warrant future investigation. Additionally, most included studies were conducted in sub-elite cohorts, which may bias results toward a greater influence of maximal strength compared to findings in elite or professional populations. Further, both absolute and relative strength measures were included, and these may have different implications for KPI relationships.

## 5 Conclusions

It is evident from the studies included in this investigation that within-game rugby union and rugby league KPIs typically have stronger links to maximal strength than to measures of speed-strength. The findings underscore the critical role of maximal strength in actions such as tackling and reducing errors, while highlighting the complementary but less dominant contribution of speed-strength to other KPIs, such as carries and line breaks. However, the analysis also reveals the nuanced and context-dependent relationships between physical qualities and specific KPIs. For instance, the influence of team-level factors, such as game model, competition grade, and player profiles, should also be considered when applying these findings. By focusing on the physical qualities most strongly linked to within-game performance, practitioners can better target resistance-training interventions, ultimately improving team and individual success in the rugby codes.

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## Declarations

**Conflicts of Interest** The authors have no conflicts of interest to report in relation to this work.

**Data Availability** The following link provides the R code to run the analysis: https://github.com/m2huynh/SPOA-D-25-00240/blob/main/Maximal%20Strength%20or%20Speed-Strength/Scripts/Script.R.

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